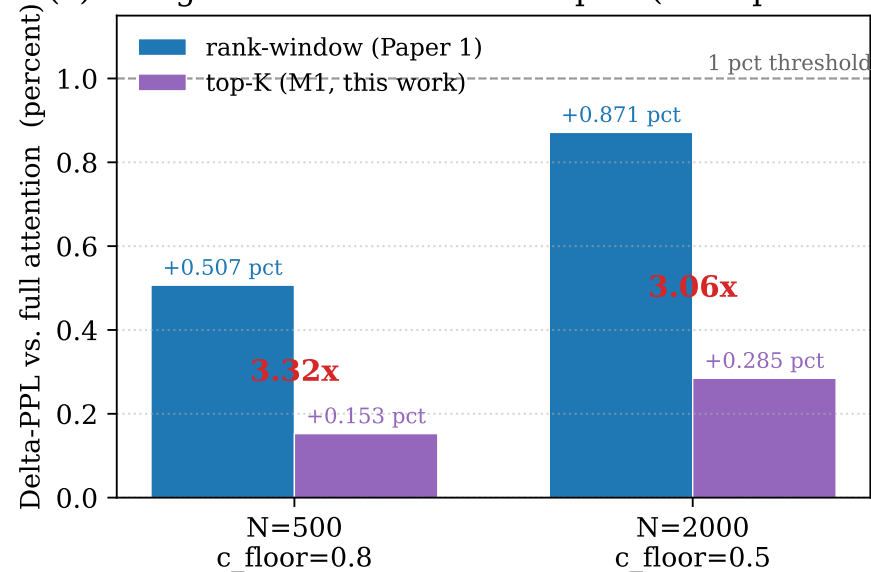


(a) M1 gate: rank-window vs. top-K (3x improvement)



(b) Absolute PPL: top-K nearly matches baseline

